

$$\Rightarrow Y_0 \cdot A = B^{-1} \text{ (Theorem 2(e))}$$

$$\Rightarrow (Y_0 \cdot A) \cdot A^{-1} = B^{-1} A^{-1} \Rightarrow Y_0 \cdot (AA^{-1}) = B^{-1} A^{-1} \text{ (Theorem 2(a))}$$

$$\Rightarrow Y_0 \cdot I_n = B^{-1} A^{-1} \text{ (By defn. of inverse)} \Rightarrow \boxed{Y_0 = B^{-1} A^{-1}} \text{ (Theorem 2(e))}$$

(Uniqueness part proved).

$\therefore B^{-1} A^{-1}$ is the unique matrix s.t. $(AB) \cdot (B^{-1} A^{-1}) = (B^{-1} A^{-1}) \cdot (AB) = I_n$.

By defn. of inverse, $\boxed{(AB)^{-1} = B^{-1} A^{-1}} \quad \square$

c) By defn. of inverse, there exists a unique matrix A^{-1} s.t. $AA^{-1} = A^{-1}A = I_n$.

(By defn. of identity matrix)

$$(AA^{-1})^T = (I_n)^T \Rightarrow (A^{-1})^T \cdot A^T = (I_n)^T \text{ (Theorem 3(d))} \Rightarrow (A^{-1})^T A^T = I_n$$

(By defn. of identity matrix)

(Existence part proved)

Let, there exist a $n \times n$ matrix B s.t. $(A^T)B = I_n$.

$(A^{-1})^T$ obviously exists. $(A^{-1})^T \cdot (A^T B) = (A^{-1})^T \cdot I_n$

$$\Rightarrow (A^{-1})^T \cdot A^T B = (A^{-1})^T \cdot I_n \text{ (Theorem 2(a))}$$

$$\Rightarrow I_n \cdot B = (A^{-1})^T \cdot I_n \text{ (Given)} \Rightarrow \boxed{B = (A^{-1})^T} \text{ (Theorem 2(e))}$$

Let, there exist a $n \times n$ matrix C s.t. $C(A^T) = I_n$.

$$(CA^T) \cdot (A^{-1})^T = I_n \cdot (A^{-1})^T \Rightarrow C(A^T \cdot (A^{-1})^T) = I_n \cdot (A^{-1})^T \text{ (Theorem 2(a))}$$

$$\Rightarrow C \cdot I_n = I_n \cdot (A^{-1})^T \text{ (Proved above)} \Rightarrow \boxed{C = (A^{-1})^T} \text{ (Theorem 2(e))}$$

(Uniqueness part proved).

$\therefore (A^{-1})^T$ is the unique matrix s.t. $(A^{-1})^T \cdot A^T = A^T \cdot (A^{-1})^T = I_n$

By defn. of inverse, $\boxed{(A^T)^{-1} = (A^{-1})^T} \quad \square$

The following generalization of Theorem 6(b) is needed later.

The product of $n \times n$ invertible matrices is invertible, and the inverse is the product of their inverses in the reverse order.